

WILLIAM NIXON

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RESEARCH INTERESTS

ML-for-systems, systems-for-ML, operating, storage and distributed systems.

EDUCATION

University of Chicago

Ph.D. Student in Computer Science. CGPA: 4.00

Advised by: Prof. Haryadi S. Gunawi and Prof. Juncheng Yang (Harvard)

Sep 2025 - Present

Chicago, Illinois

Bandung Institute of Technology

B.S. in Computer Science. CGPA: 3.95.

TOEFL: 115 [30 R, 29 L, 27 S, 29 W]

Jul 2021 - Jul 2025

Bandung, Indonesia

PUBLICATION

Ray A. Sinurat, **William Nixon**, Philip Carns, Huihuo Zheng, Sandeep Madireddy, Sam Foreman, Troy Arcomano, Robert Ross, Haryadi S. Gunawi, Hariharan Devarajan. "**SysX: Taming I/O Optimization for Deep Learning at Scale**". *In Submission*.

Hongshu Yan, Yunjia Zheng, **William Nixon**, Yazhuo Zhang, Pranav Bhandari, Haryadi S. Gunawi, Ana Klimovic, Juncheng Yang. "**Demystifying and Optimizing Slab Rebalance in Memory Caches**". *In Submission*.

Haocheng Xia, **William Nixon**, Bintang Dwi Marthen, Pranav Bhandari, Juncheng Yang. "**Learning-Augmented Heuristics: Simple yet Smart, Robust and Interpretable Cache Eviction**". *In Submission*.

Samuel Cahyawijaya, Holy Lovenia, Joel R. A. Moniz, ... **William Nixon**, ... Peerat Limkonchotiwat. "**Crowdsource, Crawl, or Generate? Creating SEA-VL, a Multicultural Vision-Language Dataset for Southeast Asia**". *Association for Computational Linguistics (ACL)*, 2025.

Holy Lovenia, Rahmad Mahendra, Salsabil M. Akbar, ... **William Nixon**, ... Samuel Cahyawijaya. "**SEACrowd: A Multilingual Multimodal Data Hub and Benchmark Suite for Southeast Asian Languages**". *Empirical Methods in Natural Language Processing (EMNLP)*, 2024.

RESEARCH EXPERIENCE

Efficient Caching

Sep 2025 – Now

- Analyzed existing **slab rebalance** strategies for slab-based memory caches, characterizing their behavior on real-world traces and their effectiveness in guiding memory allocation decisions.
- Evaluated a new **adaptive marginal-hits** strategy, demonstrating faster convergence than prior approaches and providing empirical evidence for its **near-optimality**.
- Designed a **learning-augmented caching framework** that adapts to workload patterns while preserving the simplicity and efficiency of heuristic policies.
- Implemented the framework on **S3-FIFO** in **CacheLib**, benchmarking performance and showing that **cache-level features** improve both effectiveness and interpretability.

Data Stall Mitigation

Mar 2025 – Jan 2025

Undergraduate Researcher

Remote

- Collaborated with Prof. Haryadi S. Gunawi (UChicago) and Argonne National Laboratory on mitigating I/O bottlenecks that stall deep learning training.
- Evaluated **AIO**, a methodology to pinpoint model I/O bottlenecks in HPC clusters, increase effective bandwidth, and lower optimization costs using **GPU-free simulation**.
- Led experiments for optimization of **2 of 4 real-world use cases** involving weather forecast models.

Workload Drift

Sep 2023 – Mar 2025

Undergraduate Researcher

Remote

- Collaborated with Prof. Haryadi S. Gunawi (University of Chicago) to study impact of **workload drift** in systems.
- Reproduced drift-mitigation algorithms (**Matchmaker**, **DriftSurf**, **AUE**), benchmarking them under drifting workloads.
- Integrated and evaluated these algorithms in **GL-Cache** and **LinnOS**, achieving up to **5% lower miss ratio** and cutting **tail latency**.
- Packaged the research pipeline as a reproducible **Chameleon Trove artifact** to support future research.

SeaCrowd Project

Jul 2023 – Sep 2024

Undergraduate Researcher

Bandung, Indonesia

- Participated in a joint collaboration to collect NLP datasets for **Southeast Asian languages**, which are underrepresented in the research community due to data scarcity.
- Implemented standardized dataloaders for seamless integration with **HuggingFace's platform**, facilitating public dataset use by researchers and developers.
- Reviewed and validated **community-submitted dataloaders** to ensure functionality.

TEACHING

CS144 (Intro to Systems Programming): Autumn 2025

CS230 (Intro to Operating Systems): Winter 2025

[ITB] Probability and Statistics: Spring 2024; Discrete Maths: Fall 2023; Intro to Python Spring 2023

WORK EXPERIENCE

TEDx ITB

Jan 2023 – Apr 2023

Backend Developer

Bandung, Indonesia

- Served as the sole backend developer for TEDx ITB, an annual event hosting over **200 participants**.
- Developed the **ticketing system** and implemented backend logic to ensure smooth event operations, contributing to raising **20 million IDR** in revenue.
- **Tech Stack:** Next.js (Node.js & React), Strapi CMS, SendGrid.

Bukit Vista Hospitality Services

Mar 2022 – Dec 2022

Backend Developer

Remote

- Developed a **backend calendar system** for a mobile app to allow property partners to view their properties' occupancy in a Google-calendar-like manner.
- Automated various businesses processes in **finance and HR**, reducing time otherwise consumed by up to **90%**.
- **Tech Stack:** Node.js, Python, AWS, Integromat.

TECHNICAL SKILLS

Programming Languages: C, C++, Go, Java, JavaScript, Python, SQL

Machine Learning: PyTorch, TensorFlow

Systems: Apache Spark, RabbitMQ, Redis, FEMU

Cloud Computing: Google Cloud, AWS, Chameleon Cloud

Mobile Development: Android, React Native

DevOps: GitHub CI/CD, Vercel, Docker

Misc: LaTeX, Make, CUDA, OpenMP